Image Text Enhancer: A Lightweight, Easy-to-Use, and Efficient C++ Library

Algorithm Engineering 2026 Project

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Abstract

Document enhancement plays a central role in document scanning and document analysis tasks, which are central elements of any document-dependent workflow and can be found in various applications, yet many available solutions are either not focused on document enhancement, difficult to integrate into lightweight, especially mobile, workflows or not permitted for commercial use. In this paper, we present Image Text Enhancer (ITE), a modular C++ pipeline for enhancing printed and handwritten document images. Our goal is to provide a compact and efficient solution that supports practical deployment both as a command-line tool and as a reusable library in upstream applications.

Keywords

task parallelism, WebAssembly, OpenMP, image processing, C++

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1 Introduction

1.1 Background

The extraction of textual information from images — from scanned books to even receipts captured by smartphone cameras — have become an important part of everyday life [1, 8]. Any such extraction necessarily begins with the pre-processing of recorded images through techniques such as binarisation, noise removal, and geometric transformation, a process generally referred to as *document enhancement*. Numerous tools focused on this niche have been proposed. Although these approaches feature broad capabilities, they are often associated with substantial complexity on the user's end and entail considerable software dependencies. These dependencies restrict applicability, particularly in settings where binary size is constrained, such as mobile and web platforms. This limitation assumes added importance in a context where most digital media interactions now take place on smartphones and application development is increasingly oriented toward edge-computing models. Consequently, efficient and platform-independent implementations that exploit parallelisation and other optimisation techniques are of considerable importance for reducing processing time and enhancing user satisfaction.

Table 1: Comparison of related tools.

Tool	Lightweight	Licence	Commercial
ImageMagick	✗	Own	✓
Leptonica	✓	Own	✓
Unpaper	✗	Mixed	✗

1.2 Related Work

Image enhancement and document preprocessing are covered by a wide range of open-source tools and commercial software. The scope ranges from general purpose image processing and OCR libraries to specialised tools for post-processing scanned documents.

ImageMagick [4] is a suite for image processing that is available as a command line tool and as a library. It offers functions such as format conversion, resizing, filtering, and colour correction. However, ImageMagick is not specifically designed to enhance scanned or photographed images. Similar to OpenCV, the diverse features result in a large code base.

The C-based Leptonica library [3] is a general-purpose image processing and image analysis application. It is a dependency to both Tesseract and OpenCV. Although Leptonica is relatively lightweight compared to the other libraries, much like our approach, it primarily serves as a low-level library and does not offer ready-to-use, configurable processing pipelines.

A very similar tool to our approach is Unpaper [5], which focuses on post-processing scanned documents. It offers features such as deskewing, binarisation, and noise reduction that have been developed specifically for this purpose. The tool also offers a CLI variant, with which pipeline steps can be configured and disabled. Unpaper requires the library ffmpeg, which in the standard configuration consumes >100MiB of space.

1.3 Our Contribution

Unlike the existing tools and libraries described, our approach is designed to be minimalistic, extensible, and user-friendly. Instead of being a comprehensive image processing library or OCR engine, we focus on a lightweight, modular image enhancement pipeline, focused on printed and handwritten text, written entirely in C++ with only CImg as a dependency, which is itself is lightweight has no further dependencies [7]. Image Text Enhancer (ITE) has a lightweight build footprint, requiring only three build dependencies: OpenMP, X11, and JPEG. It can be imported as a library and also as a CLI executable—with 1.7 MiB on Linux.

Compared to OpenCV and ImageMagick, our work is characterised by its deliberate focus on document enhancement and compactness. Unlike Leptonica, our pipeline is specialised in the post-processing of scanned documents and unlike Unpaper, our binary including dependencies is ≈50× smaller. In addition, our licensing model (MIT-0 and CeCILL-C) permits use in research and commercial applications. By combining minimal dependencies, a modular architecture, portability through C++ and OpenMP, and a commercially usable licence, our pipeline fills a gap between large-scale, general-purpose image processing libraries and highly specialised tools for document enhancement.

We address the question: *What are challenges in optimising a document enhancement pipeline, even when aiming to adhere to best-practices in algorithm engineering?* We systematically optimise and improve our baseline implementation and evaluate the performance gains through various experiments. In a separate proof of concept, we show that our approach can be compiled for WASM and integrated into a web application.

1.4 Outline

In section 2, we introduce the pipeline that we implement and optimise. In section 3, we describe the optimisation techniques used. These are evaluated in section 4, where we compare the performance of the pipeline under different settings to the baseline implementation. Finally, we provide an outlook on possible future work in section 5.

2 Pipeline

The pipeline is designed to be modular, enabling users to activate, deactivate and configure each stage as required, both through the CLI and the library. Some stages can intentionally not be disabled. These exceptions are marked in algorithm 1 with an **asterisk**.

Algorithm 1 Image Text Enhancer Pipeline

(1) Grayscale Conversion*
(2) Deskew
(3) Contrast Enhancement
(4) Denoise: (Adaptive) Gaussian and Median Blur
(5) Binarisation*: Otsu, Sauvola, Bataineh
(6) Morphological Operations
(7) Colour passthrough

The user can choose between different algorithms for certain stages—as these have different effects on the result, users can select the algorithm best suited to their application. For instance, Bataineh's adaptive binarisation method with automatic window adjustment [2] is helpful when binarising historical, handwritten documents with uneven writing and lighting, whereas Otsu's method [6] is a good choice for binarising typewritten documents with uniform font and lighting. The pipeline options and standard configuration of the pipeline are documented in the help print of the CLI application—the user only adds the input and output file. The standard configuration delivers good results for most use cases and are based on fixed recommendations from the authors. This means that the pipeline can be applied immediately. However, users can also customise the configurations to optimise the pipeline and improve their specific results.

We provide a command-line application that can be used independently of a program to access the pipeline. This enables the pipeline to be configured and used directly via the command line.

3 Optimisations

We hypothesise that the baseline implementation fails to scale linearly with problem size, although such scaling would be desirable for image processing workloads. We therefore concentrated on identifying low-hanging fruits at the level of high-level algorithmic design, for example by selecting more efficient algorithms for specific tasks, while aiming to adhere to best practices such as avoiding false sharing and writing code amenable to auto-vectorisation. For the purposes of the paper, we focus on the stages, whose optimisations had the biggest effect. The full list of applied optimisations is documented in the project's repository.

3.1 Deskew

Instead of using a "rotate-and-project" approach, which repeatedly rotates the image to find an appropriate angle, we applied the Radon transform, which gives an acceptable rotation after the first application. Also, we removed an overlook, where the deskew operation was applied to the grayscale and colour image, if the colour passthrough option was enabled.

3.2 Bataineh

The main feature and bottleneck of the Bataineh algorithm is the adaptive sliding window, which takes into account the number of pixels from three categories—background, foreground and confusion [2]. This requires summing over areas of the image with masks reflecting these categories. The baseline approach was to sum using a nested for loop for each window. The optimised variant uses two integral image for background and confusion pixels, yielding a significant complexity reduction. We estimate our baseline implementation with $O(n \cdot w^2)$—where n is the number of pixels and w the window size—while the improved implementation is in $O(n)$.

3.3 Contrast Enhancement

The contrast enhancement stage simply linearises the histogram. The baseline implementation was not parallelised, which was the first optimisation. Also, the lookup-table is pre-computed, which allows for auto-vectorisation in the application of the lookup-table to the image. Similarly, the Sauvola binarisation method, uses a branchless lookup in the optimised variant.

3.4 Morphology

The pipeline only supports dilation and erosion, which can both be realised by linear filters. The baseline implementation used four nested for-loops[1] for iterating the image and scanning the neighbourhood. Since the filter is separable, it is divided into a horizontal and vertical pass. Instead of scanning the neighbourhood, a queue is used to compute the min/max of a given 1D window in amortised $O(1)$.

[1] for colour images it is five, as it is done for each channel, but our approach is only used on binary images

4 Experiments

To evaluate the performance of the pipeline with the optimisations, various experiments are conducted and the results are compared with the baseline implementation. The experiments examine the runtime when processing images of different sizes with and without parallelisation.

The experiments assess three aspects: (1) single-threaded and multi-threaded runtime scaling, (2) strong-scaling behaviour and (3) profiling. The plot and profiling data is accessible online.

4.1 Setup

All stages of the pipeline are enabled, with Bataineh binarisation and the adaptive Gaussian and adaptive median filters selected.

For our measurements, we use a machine with an AMD Ryzen 9950X3D 16-core processor (32 hyperthreads) running WSL2 with Ubuntu 24.04.1 LTS with 64 GB of RAM[2]. The profiling benchmarks were conducted on native Ubuntu 24.04.1 LTS. Each core has a base frequency of 4.3 GHz, with boost frequency of around 5.5 GHz. Care has been taken to limit the number of background processes, so as not to affect performance.

We selected a small but representative set of document images spanning both document type and resolution. The benchmark includes textual documents of various sizes and degrees of detail. We will refer to the images in terms of their width in pixels, i.e. 1K or 16K (for details, see the repository). These choices aim to capture common document-enhancement scenarios while also testing scalability to very large inputs.

The pipeline was configured to apply the following operations: deskew, adaptive gaussian, simple median filter, adaptive median filter, Bataineh binarisation, despeckle, dilation, erosion and color pass. All these stages were also timed individually. The benchmarks do not report the time necessary for loading and writing the image to and from disk.

4.2 Results

Figure 1 shows runtime versus image resolution for the baseline and optimised implementations. The optimised implementation scales almost linear, surpassing the baseline. For $c_1, c_2 < 1.7 \cdot 10^{-5}$

$$t_{\text{baseline}} \propto c_1 \cdot s^{1.660}$$
$$t_{\text{optimised}} \propto c_2 \cdot s^{1.071}$$

The optimisations significantly reduce runtime, especially for larger inputs. The optimised implementation scales nearly linearly, with some overhead, i.e. serial regions. Table 2 points at two main improvements: **Deskew** and **Binarisation (Bataineh)** which are individually 99% faster. However, the runtime scaling benchmark does not tell us about the use of parallel resources. We therefore use a strong-scaling benchmark on a fixed problem size. We tested 1 to 32 threads. In Figure 2 we observe a well-behaved speedup of the baseline implementation, whereas the optimised code scales worse, where a 32-fold thread increase results in roughly 4-fold speedup. This suggests that there are serial regions in the program,

[2]It is clear that this comes with a performance penalty, however we can expect an even additive penalty across our conducted tests. See this blog post which tested the penalty on very similar hardware.

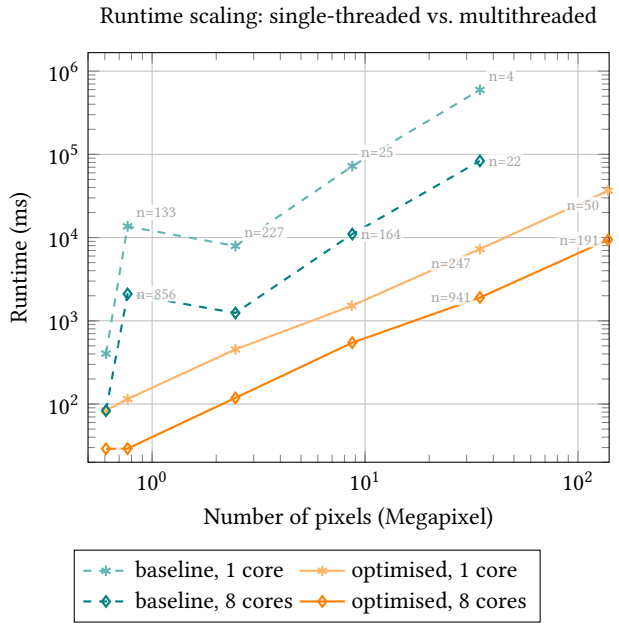

Figure 1: Runtime as a function of image size for the baseline and optimised implementations in single-threaded and 8-core execution. Dashed lines indicate 1-core measurements, solid lines indicate 8-core measurements. Labels are shown only for measurements whose sample count differs from the standard case of 1000 repetitions. For clarity, all but one documents with image size of $\approx 10^1$ Megapixels were omitted, as the results were representative for each other.

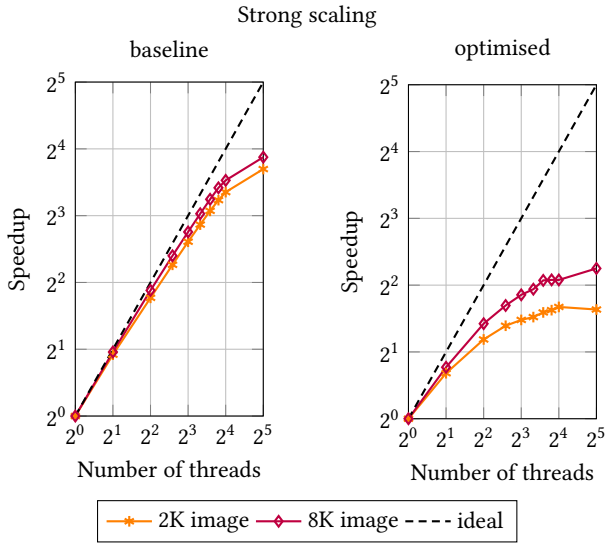

Figure 2: Strong scaling behaviour of the baseline and optimised implementations. Speedup is shown as a function of thread count for two input images (2K and 8K). The dashed black line indicates ideal linear scaling. Each point in the plot represents the mean of 50 trials.

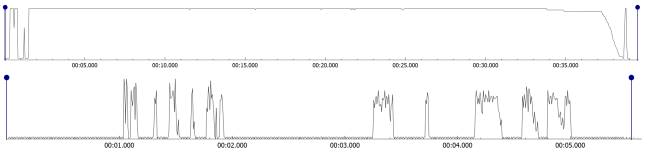

Figure 3: AMDuProf profiles of baseline (up) optimised (down) variant—number of threads over time. Profiling conducted with a 16K image and 32 cores on native Ubuntu.

that dominate after the optimisations have been applied, or, that the CPU speed is bound by waiting on e.g. main memory. It is also expected, that the scaling is slightly better for images with a larger problem size, namely because unavoidable serial regions like I/O take up less of the program's execution time.

For more information on the bottlenecks, we created a profile using AMDuProf over an 8K image (find detailed profiling reports in the repository). Figure 3 shows the profile plot of the **baseline implementation**. The plot is dominated by the Bataineh algorithm (the large plateau in the middle), meaning it makes good use of parallel resources—most time is spent on the summing of the area, which is indeed highly parallelised. This is also reflected in Table 2, where the Bataineh binarisation has 95% of total runtime attributed to it. This explains the good strong scaling performance and also why the optimised variant is so much quicker.

The **optimised profile** shows a different pattern—very few truly parallel regions can be spotted. The plot shows ≈ 27% I/O, which are excluded in the benchmarks, as well as large serial regions and memory allocations inside of pipeline stages, which can be attributed to e.g. allocation of stage-local scratch buffers for the morphological functions, masks for the Bataineh algorithm and even the output image for the binarisation functions. From the profiling and associated Assembly, we estimate that 15% of the total pipeline time (on an 8K image in a single serial profiling run) was spent on such memory operations. Another caveat are the CImg library functions—both the gaussian and median filter appear serial, even when forcing CImg to use OpenMP. The single largest impact in this benchmark scenario is the median filter, with ≈ 15.1% serial execution on an 8K image. The theoretical speedup $S_{\max}$ is limited by the serial regions of the program. Our observed speedup is however 4.76 (see Figure 2), implying, that these regions are not fully serial.

$$S_{\max} \approx \frac{1}{0.301} \approx 3.32 < 4.76 \approx \frac{1}{0.21} \approx S_{\text{observed}}$$

With this hypothesis, we introduced another optimised variant, where the caller allocates scratch and output buffers, meaning they can be reused throughout the pipeline. With this we found a speedup of 6.16 on an 8K image and 32 threads.

4.3 Conclusions

The experiments show that our hypothesis, that algorithmic improvements were important for improving the runtime, is true. Compared with the baseline, the optimised implementation achieves nearly linear runtime growth with image size and reduces total runtime by up to 97.8% on some images. However, the strong-scaling benchmark shows, that the main reason the optimised variant is faster, are the algorithmic optimisations and resulting lower work complexity. Applying the optimisations exposed new ceilings, which are for one memory related and also ineffective use of library functions. Also, the evaluation is incomplete, as only a fraction of e.g. binarisation algorithms were evaluated. Therefore, further work is necessary to make this library production ready.

Table 2: Comparison of phase runtimes (8K, 8 Cores)

Phase	baseline (ms)	optimised (ms)	diff (%)
Init & Copy	144.139	139.457	-3.248%
Grayscale	52.586	22.590	-57.042%
Deskew	**308.601**	**2.200**	**-99.287%**
Contrast	30.768	6.174	-79.934%
Adaptive Gaussian	250.973	262.134	+4.447%
Median Blur	287.138	277.698	-3.288%
Adaptive Median	883.230	499.437	-43.453%
Binarization	**81,352.114**	**503.254**	**-99.381%**
Despeckle	0.002	0.002	0.000%
Dilation	46.541	58.108	+24.853%
Erosion	103.450	56.895	-45.002%
Color Pass	12.556	12.598	+0.335%
Total	**83,472.104**	**1,840.552**	**-97.795%**

5 Future Work

The future work is settled into three categories: visual performance improvements, implementation improvements and benchmarking. For one, the visual performance of the library was not systematically evaluated in this paper, as this was not the focus of this lecture. Also, the benchmark was limited to only the Bataineh binarisation algorithm, leaving out the also implemented Otsu and Sauvola, which have very different behaviour. In future work the WASM runtime performance could be evaluated, while also aiming to improve the performance by using intrinsics or rewriting the code further.

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